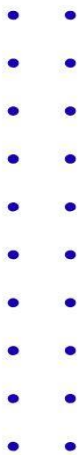


DESIGNING AND DEVELOPING
OBJECT-ORIENTED COMPUTER PROGRAMS

FITNESS TRACKER

SPRING 2024 – WINTER 2025



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Preface

This fitness tracker has user-specific fitness tracking and goal management features. User can register and log in into system for monitoring progress, setting goals and recording activities.



Fig 1 – Using fitness tracker app in laptop (freepik, 2026)

Task 1 – Program

MHKFitness application helps user track calories during exercise. It encourages users to get a healthy and happy lifestyle. It contains 6 different exercise modes to analyze daily activity. Main screen of this application is shown in figure 1.1 and original source code and its related files are shown in figure 1.2.

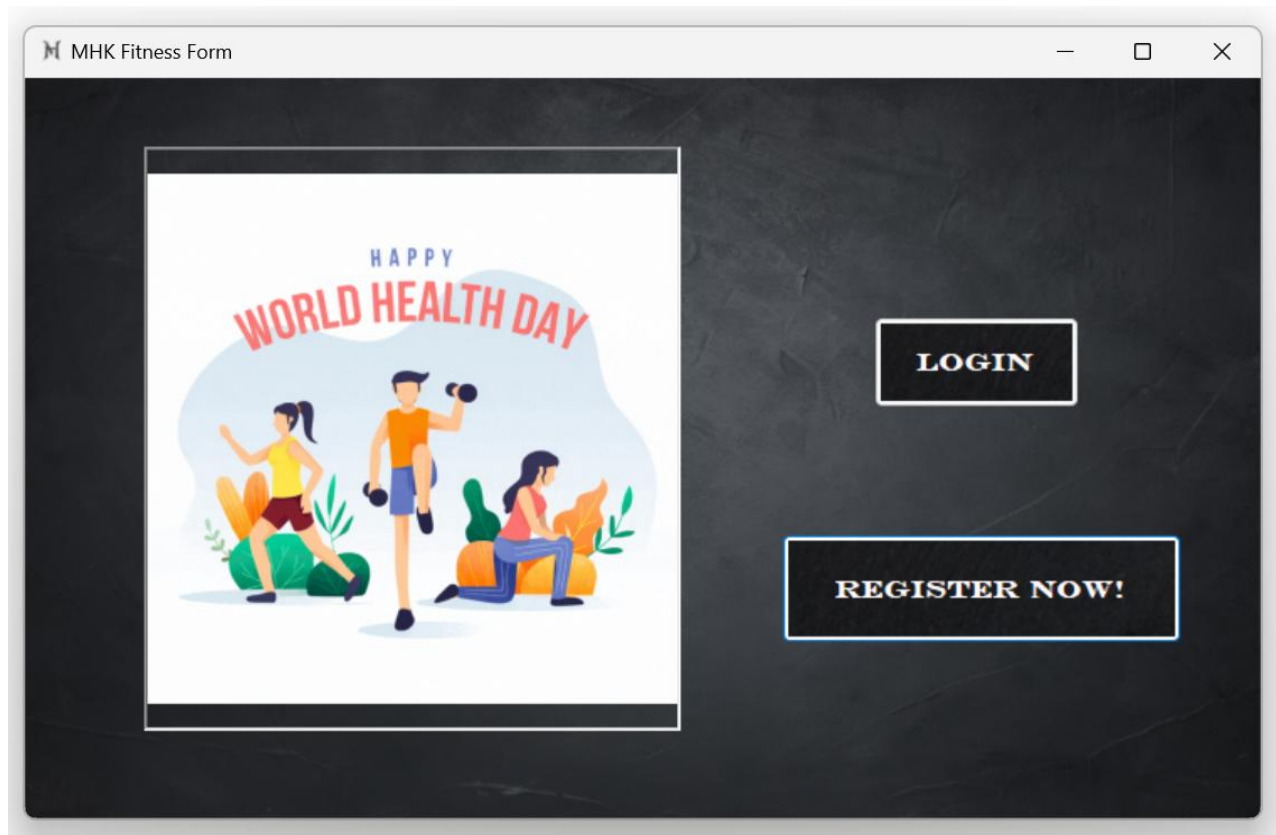


Fig 1.1 – Main screen of MHKFitness application

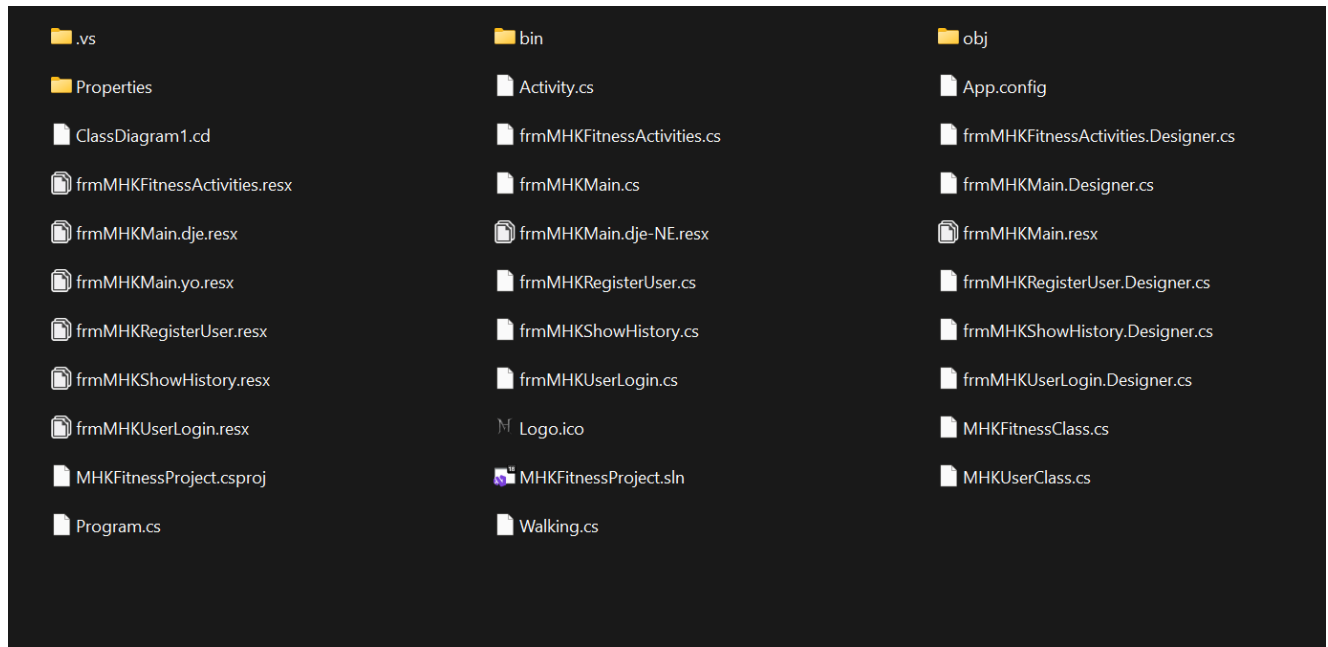


Fig 1.2 – Original source code file of MHKFitness application

Burned calories is calculated by using MET formula ($\text{Calories} = \text{MET} \times \text{body weight (kg)} \times \text{exercise time (hours)}$). MHKFitness application includes 6 types of activities such as walking, swimming, hiking, weight lifting, cycling and running. MET values used in this application are taken from ChatGPT (2026).

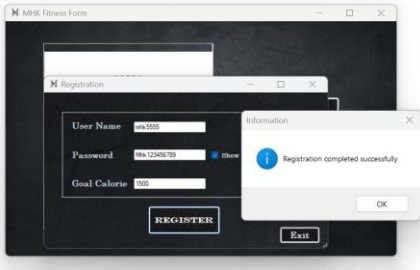
(BestCalculators, 2026) (chatgpt, 2026)

Task 2 – Testing

1. User registration

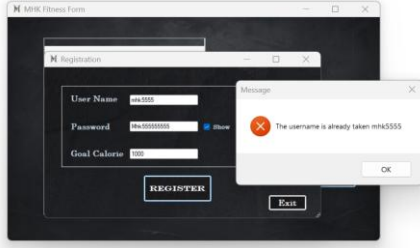
Normal testing

Username, password and goal calorie are entered and registered as normal. Then, “REGISTER NOW!” Button is clicked. The expected result is checked.

Case ID	Test Description	Predicted Result	Observed Result	Status
C0001	mhk5555 (username), Mhk123456789 (Password), 1500 (Goal calorie) are entered and then clicked “REGISTER NOW!” Button.	“Registration completed successfully” Message will be displayed.	 A screenshot of a web application window titled 'MHK Fitness Form'. It contains a registration form with fields for 'User Name' (filled with 'mhk5555'), 'Password' (filled with 'Mhk123456789'), and 'Goal Calorie' (filled with '1500'). There is a 'REGISTER' button and an 'Exit' button. An 'Information' dialog box is overlaid on the form, displaying a blue information icon and the text 'Registration completed successfully' with an 'OK' button.	Pass

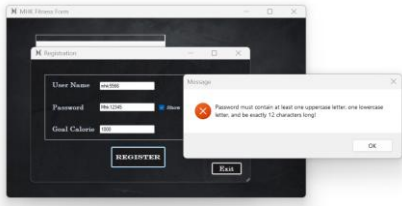
Boundary testing

Existing user name is entered and registered. The, “REGISTER NOW!” Button is clicked. The expected result is checked.

Case ID	Test Description	Predicted Result	Observed Result	Status
C0002	mhk5555 (username), Mhk5555555555 (Password), 1000 (Goal calorie) are entered and then clicked “REGISTER NOW!” Button.	“The username is already taken mhk5555” Message will be displayed.	 A screenshot of the same 'MHK Fitness Form' registration window. The 'User Name' field contains 'mhk5555'. An error message dialog box is displayed, featuring a red 'X' icon and the text 'The username is already taken mhk5555' with an 'OK' button.	Pass

Erroneous Testing

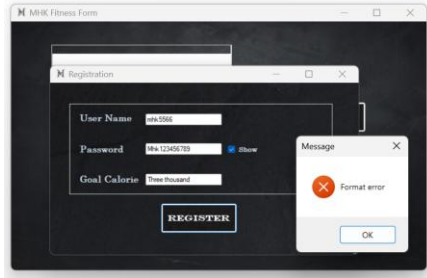
Incorrect password (Password length less than 12) is entered and registered. The, "REGISTER NOW!" Button is clicked. The expected result is checked.

Case ID	Test Description	Predicted Result	Observed Result	Status
C0003	mhk5566 (username), Mhk12345 (Password), 1000 (Goal calorie) are entered and then clicked "REGISTER NOW!" Button.	"Password must contain at least one uppercase letter, one lowercase letter, and be exactly 12 characters long!" Message will be displayed.		Pass

Incorrect password (Password length greater than 12) is entered and registered. The, "REGISTER NOW!" Button is clicked. The expected result is checked.

Case ID	Test Description	Predicted Result	Observed Result	Status
C0004	mhk5566 (username), Mhk12345678910 (Password), 1000 (Goal calorie) are entered and then clicked "REGISTER NOW!" Button.	"Password must contain at least one uppercase letter, one lowercase letter, and be exactly 12 characters long!" Message will be displayed.		Pass

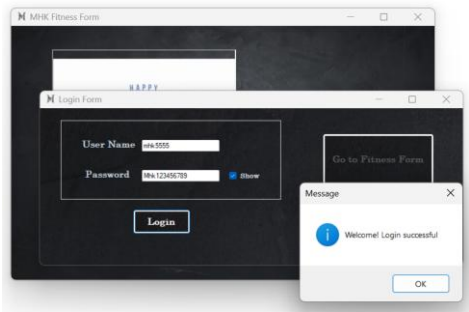
Incorrect calorie (In words) is entered and registered. Then, “REGISTER NOW!” Button is clicked. The expected result is checked.

Case ID	Test Description	Predicted Result	Observed Result	Status
C0005	mhk5566 (username), Mhk12345 (Password), Three thousand (Goal calorie) are entered and then clicked “REGISTER NOW!” Button.	“Format error” Message will be displayed.	 A screenshot of a web browser window titled 'MHC Fitness Form'. Inside, there's a 'Registration' form with fields for 'User Name' (filled with 'mhk5566'), 'Password' (filled with 'Mhk12345' and a 'show' toggle), and 'Goal Calorie' (filled with 'Three thousand'). A 'REGISTER' button is at the bottom. A 'Message' dialog box is open over the form, displaying a red 'X' icon and the text 'Format error', with an 'OK' button at the bottom.	Pass

2. User Log in

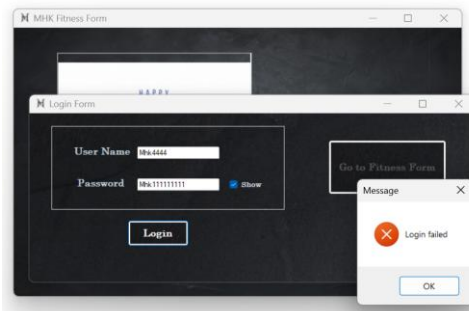
Normal testing

Valid username and password are entered and logged in. Then, "Login" Button is clicked.
The expected result is checked.

Case ID	Test Description	Predicted Result	Observed Result	Status
C0006	mhk5555 (username) and Mhk123456789 (Password), are entered and then clicked "Login" Button.	"Welcome! Login successful" Message will be displayed.		Pass

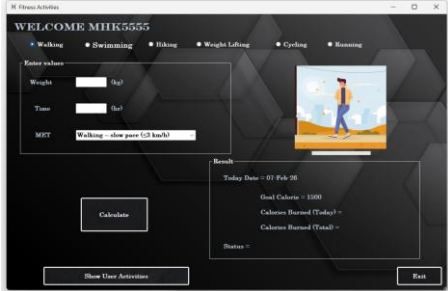
Erroneous Testing

Invalid username and password are entered and logged in. Then, "Login" Button is clicked.
The expected result is checked.

Case ID	Test Description	Predicted Result	Observed Result	Status
C0007	Mhk4444 (username) and Mhk111111111 (Password), are entered and then clicked "Login" Button.	"Login failed" Message will be displayed.		Pass

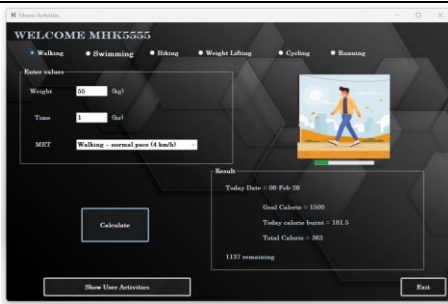
3. Fitness Form

When “Go to Fitness Form” Button is clicked. The expected result is checked.

Case ID	Test Description	Predicted Result	Observed Result	Status
C0008	When “Go to Fitness Form” Button is clicked.	The fitness activities form is displayed.		Pass

4. Walking

“Walking” exercise is chosen and weight (55kg), time (1 hour) and MET (Walking – normal pace (4km/h)) is entered. Then, “Calculate” Button is clicked. The expected result is checked.

Case ID	Test Description	Predicted Result	Observed Result	Status
C0009	Choose “Walking” exercise and enter weight (55kg), time (1 hour) and MET (Walking – normal pace (4km/h)) is entered.	Calories Burned (Today) = 181.5 Calories Burned (Total) = 363 Remaining calorie = 1137		Pass

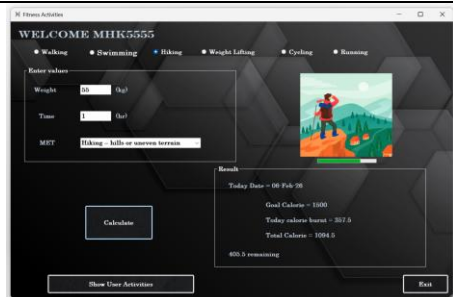
5. Swimming

“Swimming” exercise is chosen and weight (55kg), time (1 hour) and MET (Swimming – moderate effort) is entered. Then, “Calculate” Button is clicked. The expected result is checked.

Case ID	Test Description	Predicted Result	Observed Result	Status
C0010	Choose “Swimming” exercise and enter weight (55kg), time (1 hour) and MET (Swimming – moderate effort) is entered. (4km/h)) is entered.	Calories Burned (Today) = 374 Calories Burned (Total) = 737 Remaining calorie = 763		Pass

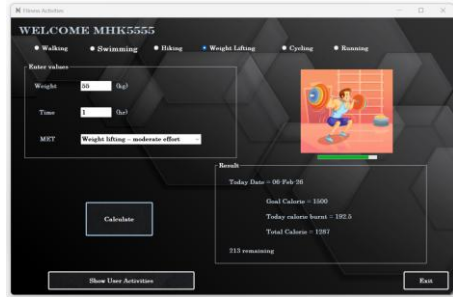
6. Hiking

“Hiking” exercise is chosen and weight (55kg), time (1 hour) and MET (Hiking – hills or uneven terrain) is entered. Then, “Calculate” Button is clicked. The expected result is checked.

Case ID	Test Description	Predicted Result	Observed Result	Status
C0011	Choose “Hiking” exercise and enter weight (55kg), time (1 hour) and MET (Hiking – hills or uneven terrain) is entered.	Calories Burned (Today) = 357.5 Calories Burned (Total) = 1094.5 Remaining calorie = 405.5		Pass

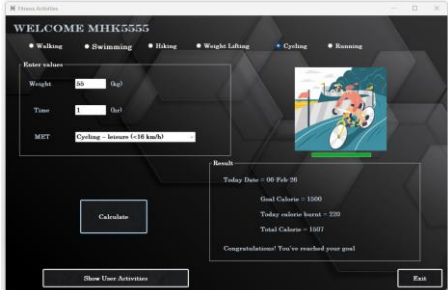
7. Weight Lifting

“Weight Lifting” exercise is chosen and weight (55kg), time (1 hour) and MET (Weighting lifting – moderate effort) is entered. Then, “Calculate” Button is clicked. The expected result is checked.

Case ID	Test Description	Predicted Result	Observed Result	Status
C0012	Choose “Weight Lifting” exercise and enter weight (55kg), time (1 hour) and MET (Weighting lifting – moderate effort) is entered.	Calories Burned (Today) = 192.5 Calories Burned (Total) = 1287 Remaining calorie = 213		Pass

8. Cycling

“Cycling” exercise is chosen and weight (55kg), time (1 hour) and MET (Cycling – leisure (<16km/h) is entered. Then, “Calculate” Button is clicked. The expected result is checked.

Case ID	Test Description	Predicted Result	Observed Result	Status
C0013	Choose “Cycling” exercise and enter weight (55kg), time (1 hour) and MET (Cycling – leisure (<16km/h) is entered.	Calories Burned (Today) = 220 Calories Burned (Total) = 1507 Congratulations ! You’ve reached your goal.		Pass

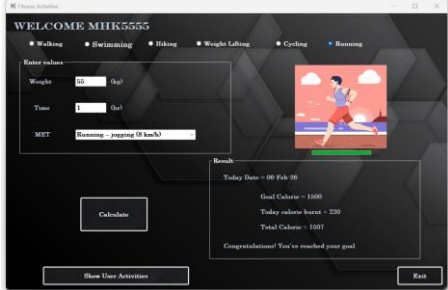
9. Running

“Running” exercise is chosen and enter weight (55kg), time (1 hour) and MET (Running – jogging (8km/h) is entered. Then, “Calculate” Button is clicked. The expected result is checked.

Case ID	Test Description	Predicted Result	Observed Result	Status
C0014	Choose “Running” exercise and enter weight (55kg), time (1 hour) and MET (Running – jogging (8km/h) is entered.	Calories Burned (Today) = 220 Calories Burned (Total) = 1507 Congratulations ! You’ve reached your goal.		Pass

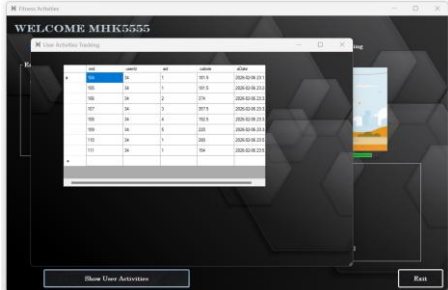
10. Reporting

When target calorie is reached, the system is displayed as “Congratulations! You’ve reached your goal.”

Case ID	Test Description	Predicted Result	Observed Result	Status
C0014	Choose “Running” exercise and enter weight (55kg), time (1 hour) and MET (Running – jogging (8km/h) is entered.	Calories Burned (Today) = 220 Calories Burned (Total) = 1507 Congratulations ! You’ve reached your goal.		Pass

11. History

When “Show User Activities” Button is clicked. The expected result is checked.

Case ID	Test Description	Predicted Result	Observed Result	Status
C0015	When “Show User Activities” Button is clicked.	The history of user activities is displayed.		Pass

Task 3 – Class diagram

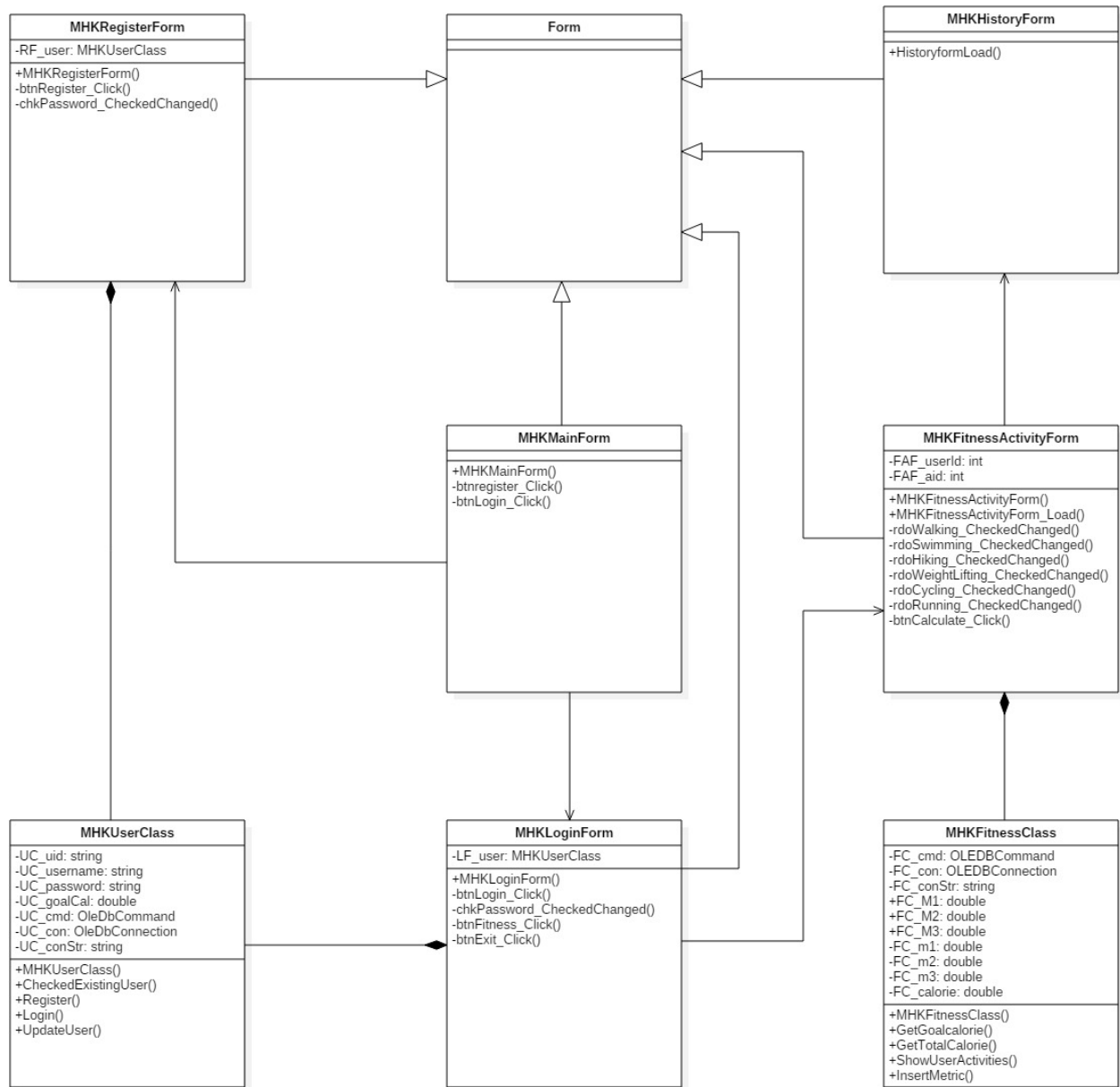


Fig 3.1 – Class diagram for MHKFitnessProject

In MHKFitnessProject, 2 types of classes are obtained such as design classes and controller classes. Design classes are also called form classes that include MHKMainForm, MHKLoginForm, MHKRegisterForm, MHKFitnessActivityForm and MHKHistoryForm. Controller classes include MHKUserClass and MHKFitnessClass.

Inheritance

It means one class is a kind of other class. Parent class and child class are present and child class inherits from parent class. So, it can extend methods for reusing behavior. (geeksforgeeks, 2025)

In MHKFitnessProject, Form class is called parent class and MHKMainForm, MHKLoginForm, MHKRegisterForm, MHKFitnessActivityForm, MHKHistoryForm are inherited from Form class (child classes).

Composition

It means one class is strongly part of other class. One class cannot exist without other which it depends. (geeksforgeeks, 2026)

In MHKFitnessProject, MHKUserClass is part of MHKRegisterForm and MHKLoginForm and MHKFitnessClass is part of MHKFitnessActivityForm.

Association

It means one class works with other class. Classes exist independently and can be one to one, one to many and many to many. (geeksforgeeks, 2026)

In MHKFitnessProject, MHKMainForm is associated with MHKRegisterForm and MHKLoginForm. MHKLoginForm is associated with MHKFitnessActivityForm. MHKFitnessActivityForm is associated with MHKHistoryForm.

Purpose of Each Class

MHKMainForm is used as welcome page that is linked with MHKLoginForm and MHKRegisterForm. MHKRegisterForm is for registering users who are not registered. The user can register with username, password and goal calorie. If users are already registered, they can directly log in by filling username and password in MHKLoginForm.

After the users are successfully logged in, they can go into MHKFitnessActivityForm. At there, they can choose activities such as swimming, walking, running, cycling, hiking and weight lifting and filling appropriate values to calculate fitness calorie and then view their total calories burnt. So, they can check if they reached their calorie goal.

If they want to know their history, they can also go MHKHistoryForm and check their previous fitness activities and calorie. MHKUserClass is connected with database to save and retrieve user data and MHKFitnessClass is connected with database to save and retrieve fitness data.

Encapsulation

Each class uses encapsulation method. Encapsulation means hiding data and controlling access. It bundles attributes and operations inside class. It uses access modifiers such as private, public and protected. Other class can access by getters and setters.

(geeksforgeeks, 2025)

Constructor

Each class has also constructor which runs automatically when form is created and is used to initialize form such as loading components, preparing UI and setting default values. For example, in MHKMainForm, MHKMainForm() is the constructor.

(geeksforgeeks, 2026)

Explanation of each class

MHKMainForm

It includes operations such as MHKMainForm(), btnLogin_Click() and btnregister_Click().

MHKRegisterForm

It includes attribute such as RF_user and operations such as MHKRegisterForm(), chkPassword_CheckedChanged() and btnRegister_Click().

MHKLoginForm

It includes attribute such as LF_user and operations such as MHKLoginForm(), chkPassword_CheckedChanged(), btnLogin_Click(), btnFitness_Click() and btnExit_Click().

MHKFitnessActivityForm

It includes attributes such as FAF_userId and FAF_aid. It has operations such as MHKFitnessActivityForm(), MHKFitnessActivityForm_Load(), rdoWalking_CheckedChanged(), rdoSwimming_CheckedChanged(), rdoHiking_CheckedChanged(), rdoWeightLifting_CheckedChanged(), rdoCycling_CheckedChanged(), rdoRunning_CheckedChanged() and btnCalculate_Click().

MHKHistoryForm

It includes operation such as HistoryformLoad().

MHKUserClass

It includes attributes such as UC_uid, UC_username, UC_password, UC_goalCal, UC_cmd, UC_con and UC_conStr. It includes operations such as MHKUserClass(), Register(), CheckedExistingUser(), UpdateUser() and Login().

MHKFitnessClass

It includes attributes such as FC_cmd, FC_con, FC_conStr, FC_M1, FC_M2, FC_M3, FC_m1, FC_m2, FC_m3 and FC_calorie. It includes operations such as MHKFitnessClass(), ShowUserActivities() , GetGoalcalorie(), InsertMetric() and GetTotalCalorie().

Conclusion

MHKFitness application for computer which has 6 fitness activities is created by visual studio. Users can set calorie goal and calculate each fitness activity's calories easily and effectively. The application is also tested by normal testing, boundary testing and erroneous Testing. So, it is well-functioned and good performance. The system detail is described as class diagram and so it reduces coding errors and improve system structure.

-----End of the Assignment-----

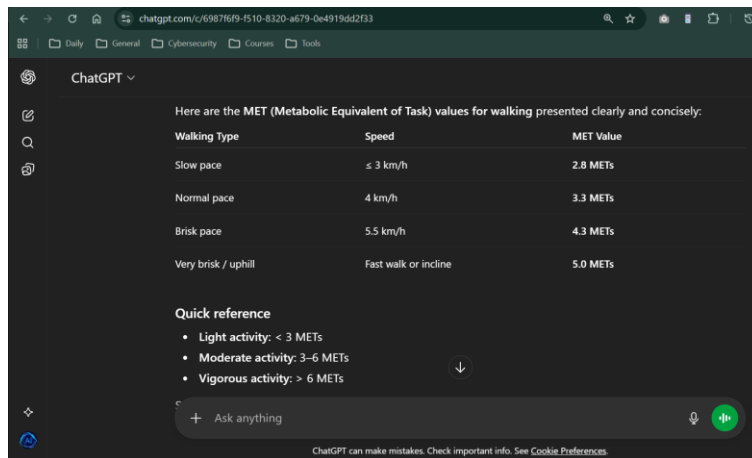
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